

CSO2 (CS3130)

# themes

automating building software

libraries, taking advantage of incremental compilation

sharing machines

multiple users/programs on one system

parallelism and concurrency

doing two+ things at once

under the hood of sockets

layered design of networks

implementing secure communication

under the hood of fast processors

caching, (hidden) parallelism, avoiding idle time

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# make

```
$ ./foo.exe
```

```
...
```

```
...
```

```
$ edit readline.c
```

```
$ make
```

```
clang -g -O -Wall -c readline.c -o readline.o
```

```
ar rcs terminal.o readline.o libreadline.a
```

```
clang -o foo.exe foo.o foo-utility.o -L. -lreadline
```

```
$
```

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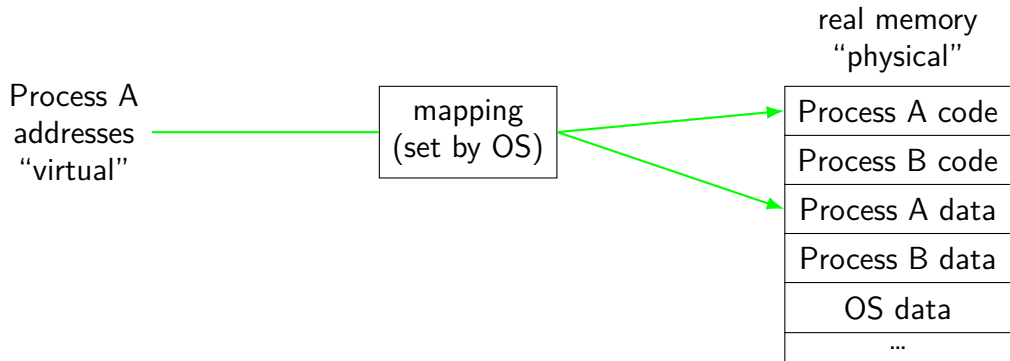
layered design of networks

implementing secure communication

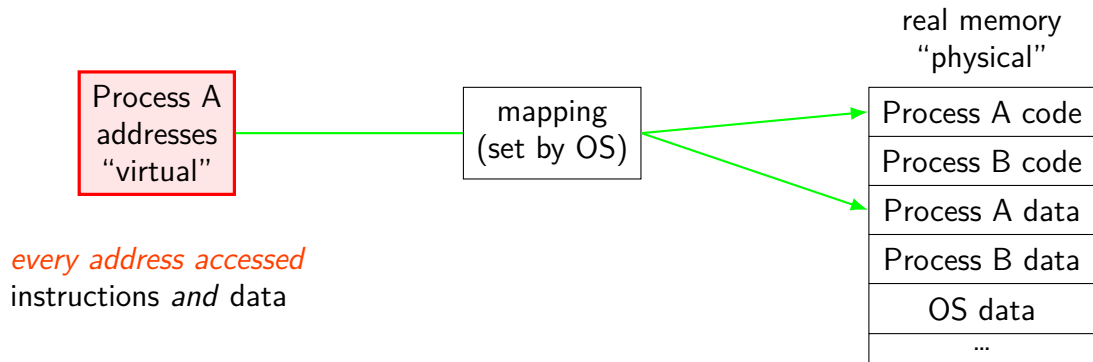
under the hood of fast processors

caching, (hidden) parallelism, avoiding idle time

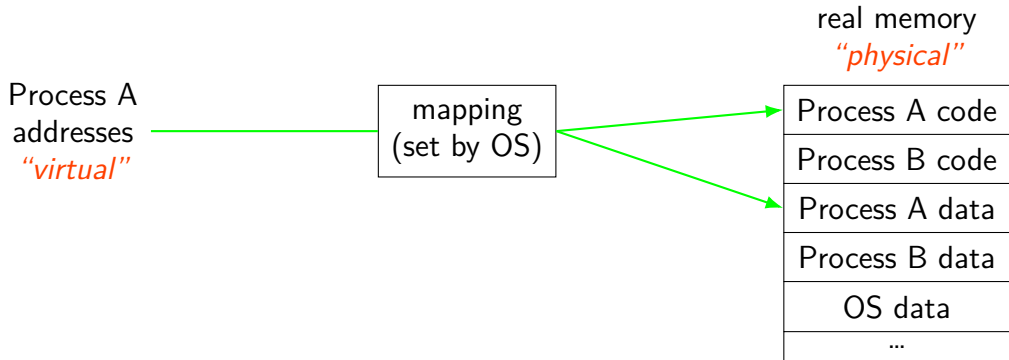
# address translation



# address translation



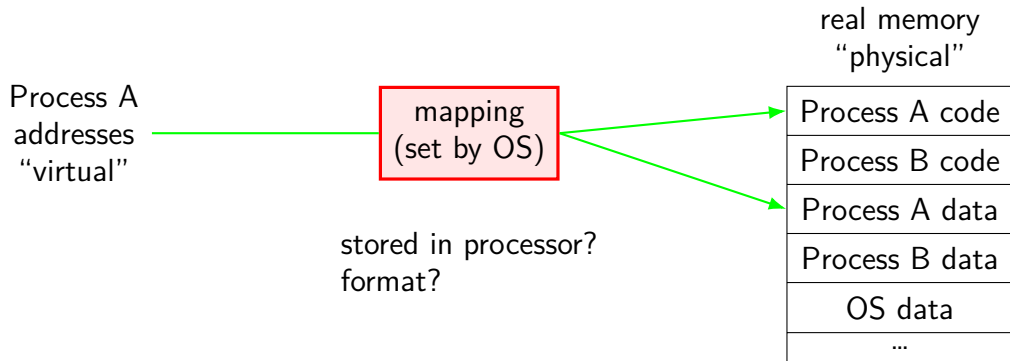
# address translation



program addresses are 'virtual'  
real addresses are 'physical'  
can be *different sizes*!

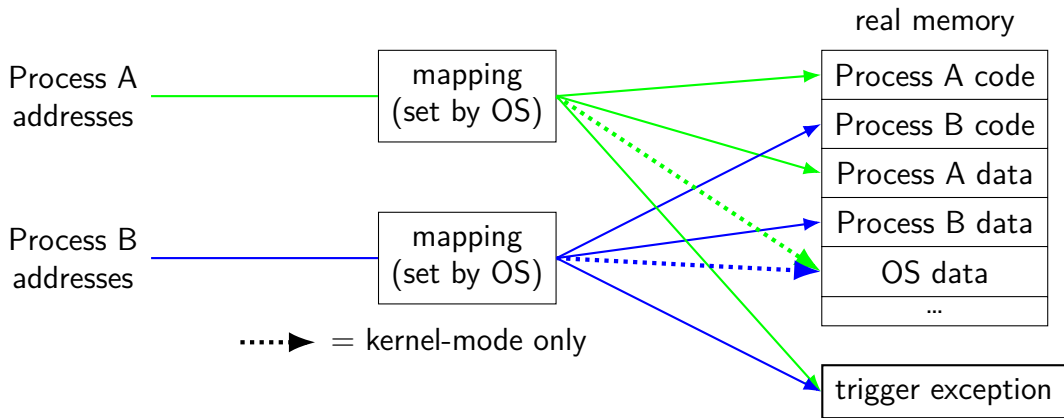


# address translation



# address spaces

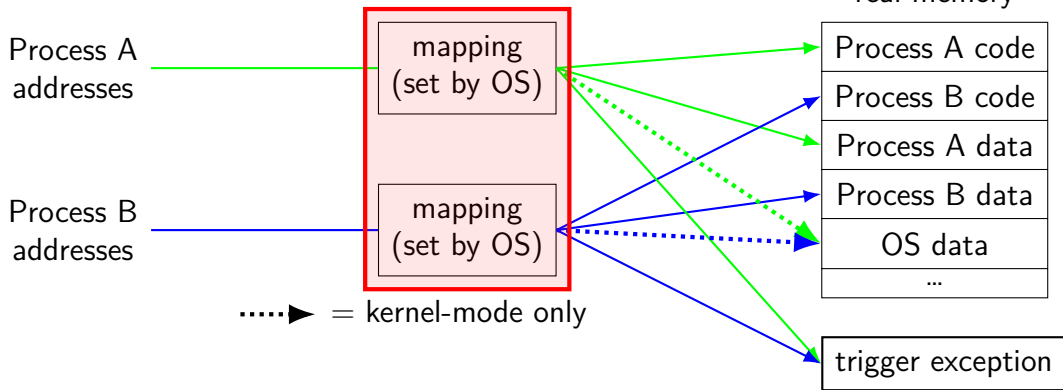
illusion of *dedicated memory*



# address spaces

illusion of *dedicated memory*

chose one during context switch



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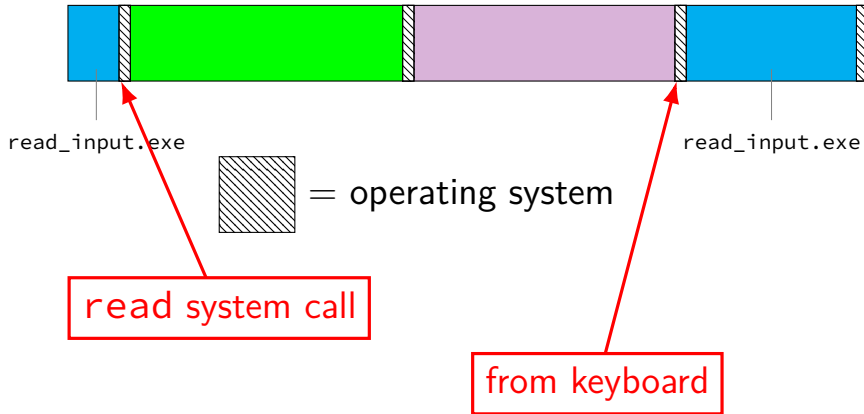
layered design of networks

implementing secure communication

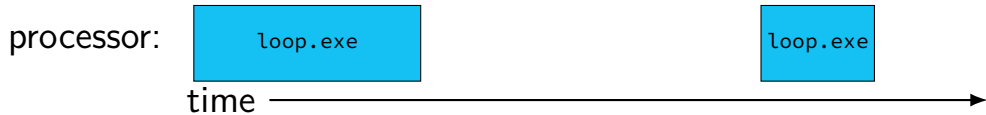
under the hood of fast processors

caching, (hidden) parallelism, avoiding idle time

# keyboard input timeline



# time multiplexing



# time multiplexing

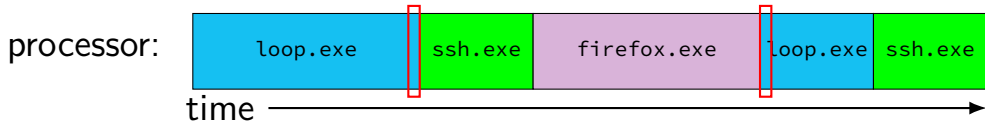


```
...  
loop: ...  
    ...  
    jmp loop  
loop: ...  
    ...
```

million cycle delay

```
    ...  
    jmp loop  
loop: ...
```

# time multiplexing



```
...  
loop: ...  
    ...  
    jmp loop  
loop: ...  
    ...
```

million cycle delay

```
    ...  
    jmp loop  
loop: ...
```

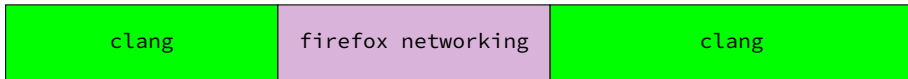


# multiple cores+threads

core 1:



core 2:



multiple cores? each core still divided up

# multiple cores+threads

core 1:



core 2:



one program with multiple *threads*

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# permissions

```
$ ls /u/other/secret  
ls: cannot open directory '/u/other/secret': Permission denied  
$ shutdown  
shutdown: Permission denied
```

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# layers

|             |                      |                                                         |
|-------------|----------------------|---------------------------------------------------------|
| application | HTTP, SSH, SMTP, ... | application-defined meanings                            |
| transport   | TCP, UDP, ...        | reach      correct      program,<br>reliability/streams |
| network     | IPv4, IPv6, ...      | reach      correct      machine<br>(across networks)    |
| link        | Ethernet, Wi-Fi, ... | coordinate shared wire/radio                            |
| physical    | ...                  | encode bits for wire/radio                              |

# layers terminology

|             |                                                         |                    |
|-------------|---------------------------------------------------------|--------------------|
| application | application-defined meanings                            |                    |
| transport   | reach      correct      program,<br>reliability/streams | segments/datagrams |
| network     | reach      correct      machine<br>(across networks)    | packets            |
| link        | coordinate shared wire/radio                            | frames             |
| physical    | encode bits for wire/radio                              |                    |

# names and addresses

| name                                 | address                               |
|--------------------------------------|---------------------------------------|
| <i>logical identifier</i>            | <i>location/how to locate</i>         |
| variable counter                     | memory address 0x7FFF9430             |
| DNS name www.virginia.edu            | IPv4 address 128.143.22.36            |
| DNS name mail.google.com             | IPv4 address 216.58.217.69            |
| DNS name mail.google.com             | IPv6 address 2607:f8b0:4004:80b::2005 |
| DNS name reiss-t3620.cs.virginia.edu | IPv4 address 128.143.67.91            |
| DNS name reiss-t3620.cs.virginia.edu | MAC address 18:66:da:2e:7f:da         |
| service name https                   | port number 443                       |
| service name ssh                     | port number 22                        |



# secure communication?

how do you know who your socket is to?

who can read what's on the socket?

what can you do to restrict this?

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# 2004 CPU

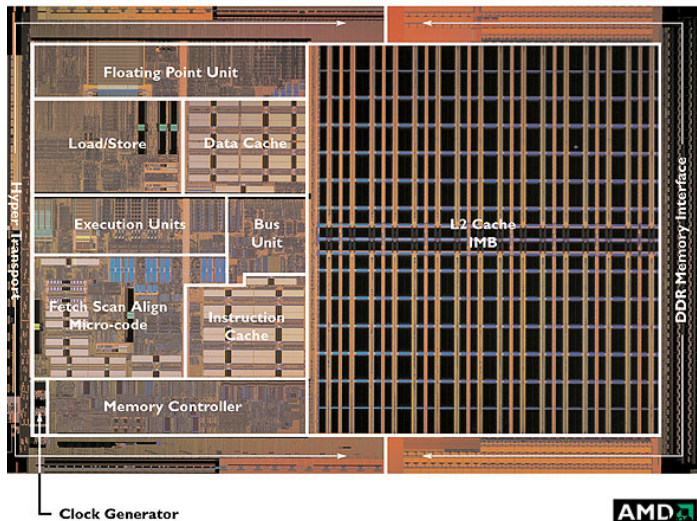


Image: approx 2004 AMD press image of Opteron die;  
approx register location via [chip-architect.org](http://chip-architect.org) (Hans de Vries)

# 2004 CPU

▲ Registers

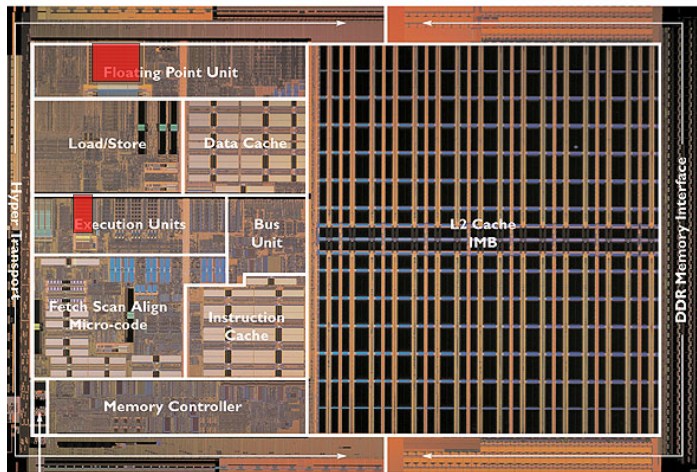


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# 2004 CPU

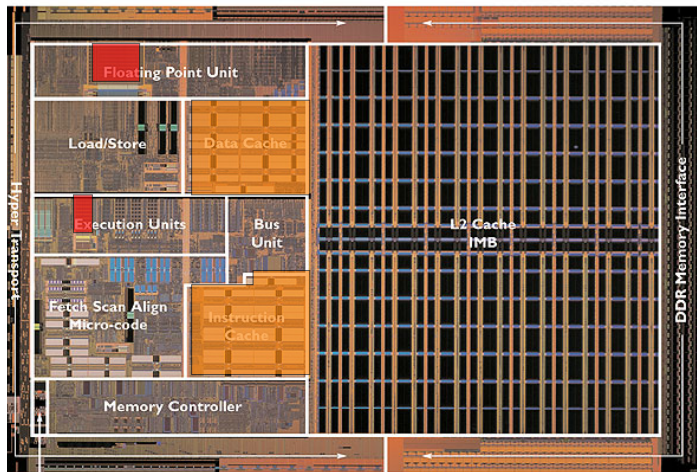
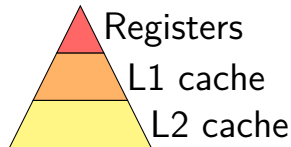
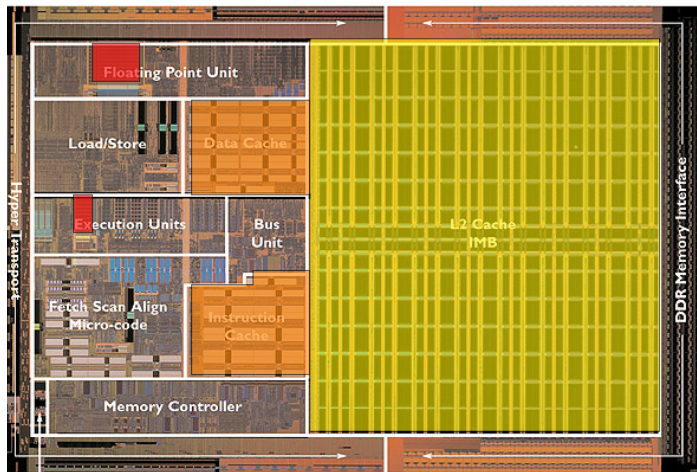
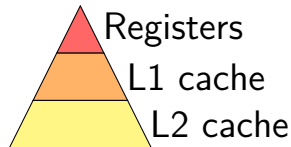
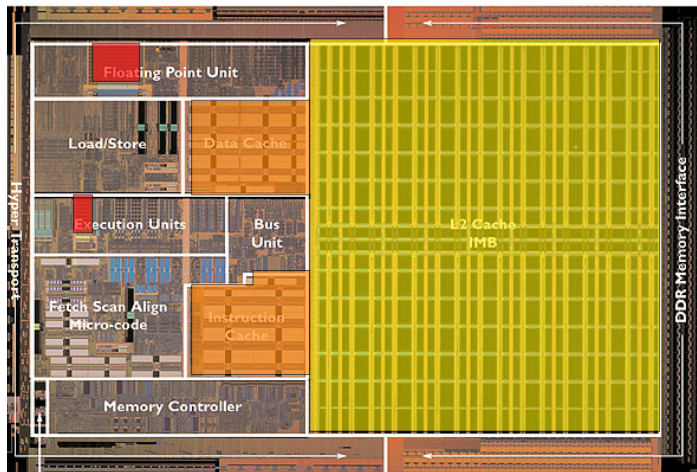


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# 2004 CPU



# 2004 CPU



# 2004 CPU

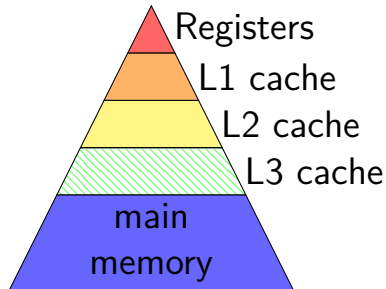
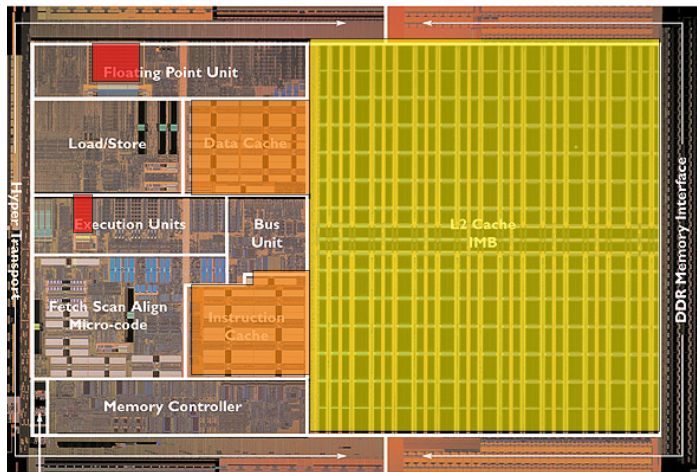
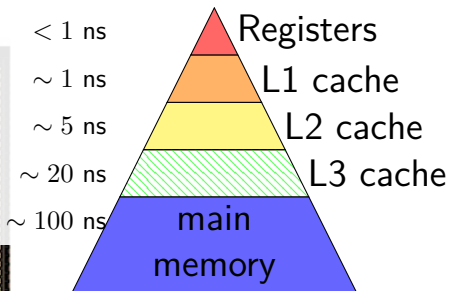
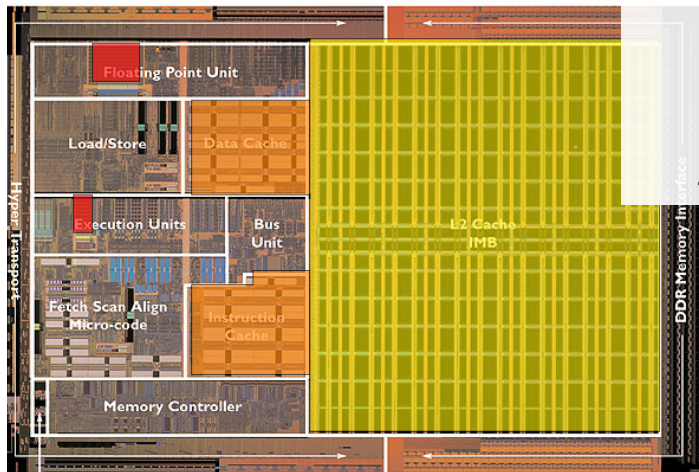


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# 2004 CPU



## some performance examples

example1:

```
    movq $1000000000000, %rax
loop1:
    addq %rbx, %rcx
    decq %rax
    jge loop1
    ret
```

about 30B instructions

my desktop: approx 2.65 sec

example2:

```
    movq $1000000000000, %rax
loop2:
    addq %rbx, %rcx
    addq %r8, %r9
    decq %rax
    jge loop2
    ret
```

about 40B instructions

my desktop: approx 2.65 sec

## some performance examples

example1:

```
    movq $1000000000000, %rax
loop1:
    addq %rbx, %rcx
    decq %rax
    jge loop1
    ret
```

*about 30B* instructions

my desktop: approx *2.65 sec*

example2:

```
    movq $1000000000000, %rax
loop2:
    addq %rbx, %rcx
    addq %r8, %r9
    decq %rax
    jge loop2
    ret
```

*about 40B* instructions

my desktop: approx *2.65 sec*

## C exercise

```
int array[4] = {10,20,30,40};  
int *p;  
p = &array[0];  
p += 2;  
p[1] += 1;
```

array =

- |                             |                  |
|-----------------------------|------------------|
| A. compile or runtime error | B. {10,20,30,41} |
| C. {10,20,32,41}            | D. {10,21,30,40} |
| E. {12,21,30,40}            | F. none of these |

## C exercise (2)

```
int *array2[4]; int array1[4] = {10,20,30,40};  
void mystery(int **p) {  
    *p = &array1[2];  
}  
int main() {  
    int **q;  
    q = array2;  
    mystery(q);  
    array1[1] = *q;  
    ...  
}
```

array1 =

- |                             |                  |
|-----------------------------|------------------|
| A. compile or runtime error | B. {10,10,30,40} |
| C. {10,30,30,40}            | D. {10,10,20,30} |
| E. {10,20,10,20}            | F. none of these |

## C exercise (2)

```
int *array2[4]; int array1[4] = {10,20,30,40};
void mystery(int **p) {
    *p = &array1[2];
}
int main() {
    int **q;
    q = array2;
    mystery(q);
    array1[1] = *q;
    ...
}
```

array1 =

- |                             |                  |
|-----------------------------|------------------|
| A. compile or runtime error | B. {10,10,30,40} |
| C. {10,30,30,40}            | D. {10,10,20,30} |
| E. {10,20,10,20}            | F. none of these |

# some avenues for review

review CSO1 stuff

labs 9–12 (of last Fall)

<https://researcher111.github.io/uva-cso1-F23-DG/>

exercises we've used in the past:

implement strsep library function

implement conversion from dynamic array to linked list

## some pointer stuff

0x040

0x038

0x030

0x028

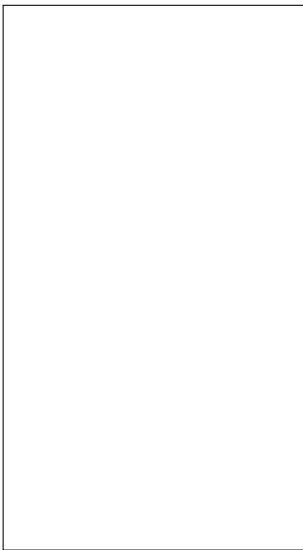
0x020

0x018

0x010

0x008

0x000



```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```



## some pointer stuff

0x040

0x038

0x030

0x028

0x020

0x018

0x010

0x008

0x000

|                |
|----------------|
|                |
| array[2]: 0x67 |
| array[1]: 0x45 |
| array[0]: 0x12 |
| single: 0x78   |
| ptr = ???      |
|                |
|                |
|                |
|                |
|                |
|                |
|                |
|                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
|       | single: 0x78   |
| 0x028 | ptr = ???      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

~~\*ptr = 0xAB;~~ runtime error

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
|       | single: 0x78   |
| 0x028 | ptr: 0x28      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

```
ptr = &single;
```

```
ptr = (int*) 0x28;    addr. of single
```

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
| 0x036 | array[1]: 0x45 |
| 0x034 | array[0]: 0x12 |
| 0x032 | single: 0x78   |
| 0x030 | ptr: 0x28      |
| 0x028 |                |
| 0x026 |                |
| 0x024 |                |
| 0x022 |                |
| 0x020 |                |
| 0x018 |                |
| 0x016 |                |
| 0x014 |                |
| 0x012 |                |
| 0x010 |                |
| 0x008 |                |
| 0x006 |                |
| 0x004 |                |
| 0x002 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

```
ptr = &single;  
ptr = (int*) 0x28;  addr. of single
```

~~ptr = 0x28; compile error~~

~~ptr = (int\*) single;~~

pointer to unknown place

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
| 0x028 | single: 0xFF   |
|       | ptr: 0x28      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;  
ptr = &single;
```

*\*ptr = 0xFF;*

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
|       | single: 0x78   |
| 0x028 | ptr: 0x2C      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

```
ptr = array;  
ptr = &array[0];  
ptr = (int*) 0x2C;
```

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
|       | single: 0x78   |
| 0x028 | ptr: 0x2C      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

```
ptr = array;  
ptr = &array[0];  
ptr = (int*) 0x2C;
```

~~ptr = array[0]; compile error~~

~~ptr = (int\*) array[0];~~

pointer to unknown place

## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0xFF |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
|       | single: 0x78   |
| 0x028 | ptr: 0x2C      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;  
ptr = &array[0];
```

```
ptr[2] = 0xFF;  
*(ptr + 2) = 0xFF;
```

```
int *temp1; temp1 = ptr + 2;  
*temp1 = 0xFF;
```

```
int *temp2; temp2 = &ptr[2];  
*temp2 = 0xFF;
```



## some pointer stuff

|       |                |
|-------|----------------|
| 0x040 |                |
| 0x038 | array[2]: 0x67 |
|       | array[1]: 0x45 |
| 0x030 | array[0]: 0x12 |
| 0x028 | single: ...    |
|       | ptr: 0x2C      |
| 0x020 |                |
| 0x018 |                |
| 0x010 |                |
| 0x008 |                |
| 0x000 |                |

```
int array[3]={0x12,0x45,0x67};  
int single = 0x78;  
int *ptr;
```

```
void change_arg(int *x) {  
    *x = compute_some_value();  
}  
...  
change_arg(&single);
```

# backup slides